Pre-registration T6: one closed set under rotation

integral octonions over the Hamming labelings

Written by Claude (Opus 5.5) at Pedro Garcia's request. Registered before any code for T6 was run.

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Object. Octonions with basis $e_0 = 1, e_1, \ldots, e_7$ and the fixed table $e_i e_{i+1} = e_{i+3}$ (indices mod 7, on 1..7), i.e. lines $(1,2,4), (2,3,5), (3,4,6), (4,5,7), (5,6,1), (6,7,2), (7,1,3)$, cyclic order positive, $e_i^2 = -1$, distinct imaginaries anticommuting. A *labeling* is a Steiner system $S(3,4,8)$ on $\{0,\ldots,7\}$ (14 blocks; there are 30). Its candidate unit set U has 240 elements: the 16 units $\pm e_a$, and for each block $\{a,b,c,d\}$ the 16 elements $\frac{1}{2}(\pm e_a \pm e_b \pm e_c \pm e_d)$. The *aligned* labeling has blocks $\{0\} \cup L$ for each line L and their complements.

Declarations. All coordinates carried as integers scaled by 2. No floats. A product is *inside* iff it equals an element of U exactly.

Predictions and kills.

H1 Closure. At least one labeling gives U closed: all $240^2 = 57{,}600$ products inside. *Kill:* no labeling closes.

H2 The aligned labeling is *not* closed. (Prior from the historical record: Kirmse's 1924 integers, as reported by Coxeter and by Conway–Smith; a recall, not a derivation.) *Kill:* it closes.

H3 Rotation. For a closed U, left multiplication by each unit permutes U exactly, and $N(uv) = N(u)N(v)$ holds exactly on all pairs. *Kill:* any unit fails to permute U.

H4 Count. Exactly 7 of the 30 labelings close. Low-confidence prior. *Kill:* any other number.

E1 Exploratory, no prediction: for each non-closed U, where the fallout products land (inside another labeling's U; half-integer coordinates; quarter-integer coordinates).

Outcomes are appended below this line and never edited into the text above.

Outcomes (appended after the run; registration text above unchanged, SHA-256 of the registered source `b3166908...6fbd`).

H1 **Holds.** 7 of the 30 labelings give a closed set: all 57,600 products inside.

H2 **Holds.** The aligned labeling is not closed: 43,008 of 57,600 products fall outside.

H3 **Holds.** In a closed set every left multiplication by a unit permutes all 240 exactly; $N(uv) = N(u)N(v)$ exactly on all pairs.

H4 **Holds.** Exactly 7 of 30 close. The prior was marked low-confidence.

E1 Across all non-closed labelings, fallout products either land exactly inside another labeling's unit set (387,072) or leave the half-integer lattice with quarter-integer coordinates (688,128). No product lands at half-integer coordinates outside every labeling. For the aligned labeling, all 43,008 fallout products are half-integer and every one lands inside another labeling's set.

Sanity checks run first: the table is a composition algebra and left-alternative on 300 random integer pairs.